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High performance B-spline multimode waveguide bends in lithium niobate on insulator

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Abstract: Lithium niobate on insulator (LNOI) has emerged as a transformative platform for integrated photonics, combining the exceptional material properties of lithium niobate with the compactness and scalability of modern photonic technologies. While traditional waveguide bend designs for LNOI face challenges such as bending losses, mode mismatch, and fabrication complexity, this study introduces an approach leveraging B-spline curves for multimode waveguide bend optimization. B-spline curves offer unparalleled design flexibility, enabling precise control of curvature profiles, smooth transitions, and simultaneous optimization of insertion loss (IL) and mode crosstalk (CT). Experimental results demonstrate that 90° B-spline-based waveguide bends achieve ultra-low insertion losses of 0.05 dB, 0.10 dB, and 0.29 dB for TE_0, TE_1, and TE_2 modes, respectively, with crosstalk below -16.71 dB across all modes for cascaded bends. These results suggest that B-spline-based designs hold promise for enabling high-performance multimode waveguides, offering a potential solution to key challenges in LNOI photonic integrated circuits.

1. Introduction

Lithium niobate (LN) has emerged as a highly competitive material for integrated photonics, owing to its wide transparency window, strong electro-optic effect, and high nonlinear optical coefficient [1–3]. These advantages place LN ahead of conventional materials such as silicon [4–6], silicon nitride [7–9], and indium phosphide [10,11]. However, traditional bulk LN devices face limitations in terms of integration density and fabrication scalability, primarily due to their relatively large device footprints and the challenges associated with precision microfabrication. The advent of lithium niobate on insulator (LNOI) technology has revolutionized the field, enabling the development of thin-film LN devices that combine the superior material properties of LN with the compactness and integration capabilities of modern photonic platforms [12–14]. LNOI offers enhanced optical confinement, reduced power consumption, and compatibility with advanced nanofabrication techniques, making it a highly promising candidate for high-performance photonic integrated circuits (PICs) [15]. This technological leap has spurred the creation of a diverse range of high-performance integrated photonic structures and devices, including electro-optic modulators [16–20], optical frequency comb generators [21–23], microring resonators [24–26], couplers [27–30], interleavers [31,32], frequency shifters [33,34], rare earth ion-doped waveguides [35,36], mode (de)multiplexers [37,38], and so forth.

In the quest for PICs that mimic electronic integrated circuits by transmitting and processing optical signals within compact footprints without compromising performance, waveguide bending is an inevitable design challenge. To date, several approaches have been proposed for designing bent waveguides, including circular bends [39], lateral offset [40], subwavelength gratings

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[41,42], inverse design [43,44], Euler bends [45,46], free-form curves (FFCs) [47] and Bézier bends [48,49]. Circular bends are straightforward to fabricate; however, they typically suffer from significant bending losses, particularly in multimode waveguides. Lateral offsets can mitigate mode mismatch between straight and bent waveguides, though they depend heavily on high-precision fabrication. Subwavelength gratings and inverse design approaches can effectively reduce the bend radius, but they introduce additional scattering losses and impose stringent fabrication requirements. Euler bends, designed to minimize curvature variation and reduce bending losses, cannot effectively address the problem of mode mismatch caused by variations in the effective refractive index during bending in anisotropic materials. Bézier bends provide enhanced control over the bending profile but are constrained in their ability to simultaneously optimize both insertion loss (IL) and mode crosstalk due to their nature as globally controlled curves. While these existing approaches address specific aspects of waveguide bend design, they often involve trade-offs between fabrication complexity, bending losses, and mode crosstalk. To overcome these limitations, a design method is required that combines flexibility, smooth curvature transitions, and the ability to optimize multiple performance metrics simultaneously.

B-spline curves offer several distinct advantages over traditional waveguide bend design approaches. Their inherent design flexibility allows precise control of the curvature profile through adjustable control points [50], enabling the simultaneous optimization of insertion loss and mode crosstalk. Furthermore, B-spline curves provide smooth curvature transitions, which help minimize scattering losses and reduce mode mismatch, particularly in multimode waveguides [51]. The local control properties of B-splines also make them highly adaptable for addressing complex design constraints, enhancing their suitability for advanced photonic integrated circuits, especially by providing greater design freedom and potential solutions for waveguide design in anisotropic materials, which is difficult to achieve with Euler curves due to their lack of local control properties. Despite these benefits, the application of B-spline curves in the design of bent waveguides for LNOI platforms has not yet been thoroughly investigated. Considering the critical importance of achieving high-performance multimode waveguides for LNOI-based PICs, a laudable goal would be to explore the design and optimization of B-spline-based bent waveguides for LNOI.

In this paper, we propose an approach for designing multimode waveguide bends on an LNOI platform utilizing B-spline curves [52]. Compared to traditional design methods, B-spline curves provide enhanced flexibility in shaping the bend profile, enabling better optimization of bending radius, insertion loss, and mode crosstalk. Our experimental results demonstrate that a 90° B-spline-based waveguide bend achieves ultra-low insertion losses of 0.05 dB, 0.10 dB, and 0.29 dB for the TE_0, TE_1, and TE_2 modes, respectively. Additionally, the crosstalk for cascaded 90° bends remains below -16.71 dB across all three modes. This low-loss waveguide bend provides an effective solution for achieving low-loss multimode transmission in LNOI-based PICs.

2. Principle and design

We selected the X-cut LNOI platform for designing and fabricating low-loss multimode waveguide bends. The X-cut LNOI wafer features a 600 nm thick LN core layer, and with the chosen manufacturing process, the slab thickness h_s was set to 300 nm, while the waveguide sidewall angle θ is approximately 70°. Figure 1(a). illustrates the 90° waveguide bend designed using B-spline curves, while Fig. 1(b). depicts the cross-section of the LN waveguide. In conventional approaches, most waveguide bends are optimized based on a fixed waveguide width. However, in anisotropic materials such as LN, the effective refractive index of the quasi-TE mode varies continuously along the bend [53]. Under these conditions, a more effective method to minimize loss involves independently optimizing the inner and outer curves of the waveguide bend. To achieve this, we employed B-spline curves, which offer a higher degree of freedom compared to

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Euler and Bézier curves, the latter two having been widely used in previous studies for bend loss optimization. The equation of the B-spline curve is given by.

$$P(u) = \sum\nolimits_{i=0}^{n} P_i N_{i,k}(u), u \in [u_{k-1}, u_{n+1}] \tag{1}$$

where P_{i} is the point on the curve and $N_{(i,k)}(u)$ is the B-spline basis function of degree k. In general, the B-spline basis function is defined by de Boor-Cox recursion and can be expressed as:

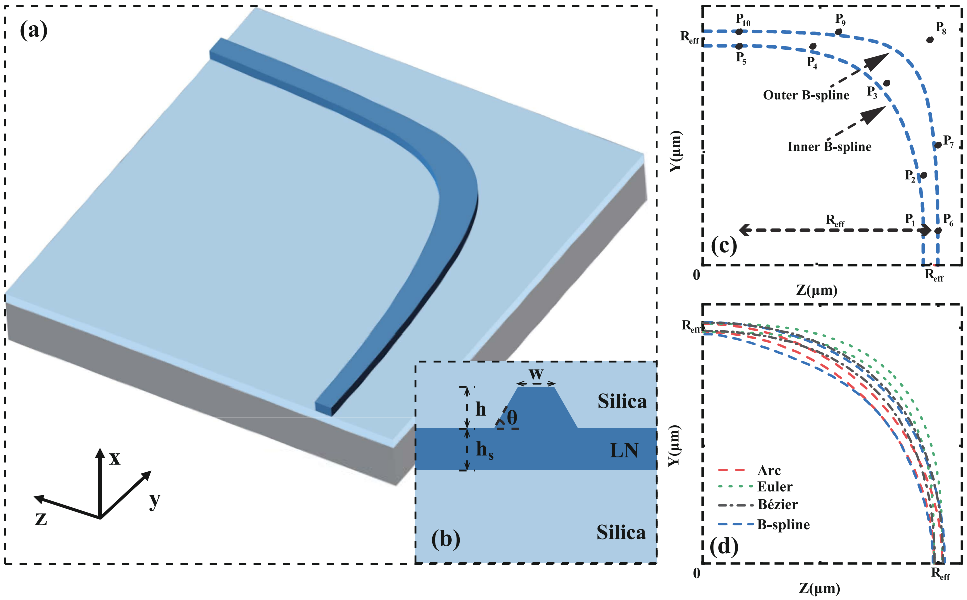

Fig. 1. (a) The schematic of the B-spline multimode waveguide bend. (b) Cross-section of the B-spline multimode waveguide bend, (c) The structural design diagram of the B-spline multimode waveguide bend, (d) the structural comparison of three commonly used curves for waveguide bends.

$$N_{i,0}(u) = \begin{cases} 1, & \mathit{if}\ u_i \le u < u_{i+1} \\ 0, & \mathit{otherwise} \end{cases} \tag{2}$$

$$N_{i,k}(u) = \frac{(u - u_i)N_{i,k-1}}{u_{i+k} - u_i} + \frac{(u_{i+k+1} - u)N_{i+1,k-1}(u)}{u_{i+k+1} - u_{i+1}} \tag{3}$$

As shown in Fig. 1(c), we selected two different sets of B-spline curves to design the 90° waveguide bend. Points P_1 (R_{eff} - $W/2$, 0) and P_6 (R_{eff} + $W/2$, 0) are the starting points of the inner and outer sides, respectively. Where R_{eff} is the effective radius of the multimode waveguide bend, and W is the waveguide width at the start and end points of the multimode waveguide bend. Points P_5 (0, R_{eff} - $W/2$) and P_{10} (0, R_{eff} + $W/2$) are the endpoints of the inner and outer sides, respectively. The control points for the inner and outer B-spline curves are P_2 [R_{eff} - $W/2$, A (R_{eff} - $W/2$)], P_3 [B (R_{eff} - $W/2$), B (R_{eff} - $W/2$)], P_4 [C (R_{eff} - $W/2$), R_{eff} - $W/2$], P_7 [R_{eff} + $W/2$, D (R_{eff} + $W/2$)], P_8 [E (R_{eff} + $W/2$), E (R_{eff} + $W/2$)], and P_9 [F (R_{eff} + $W/2$), R_{eff} + $W/2$], respectively. The adjustment of these control points is intended to modify the shape of the B-spline curve, thereby achieving the objective of reducing the transmission loss of the waveguide bend. To simplify the optimization process, certain constraints are applied to the control points: the x-coordinates of control points P_2 and P_7 are fixed, allowing adjustments only to their y-coordinates; control points P_3 and P_8 are adjusted along the line $y = x$; and control

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points P_4 and P_9 have fixed y-coordinates, with adjustments limited to their x-coordinates. The optimization ranges for each parameter are determined based on the specified radius and width of the waveguide in the given design scenario. Consequently, six parameters (A, B, C, D, E and F) need to be optimized. Since individually scanning each parameter would produce an overwhelming volume of data, the Equilibrium Optimizer (EO) algorithm is employed to achieve efficient and rapid optimization [54]. Figure 2. shows the convergence process of the B-spline waveguide bends optimized using the EO algorithm under three different radius. The number of iterations was set to 50, and the number of particles was set to 10. The optimization results for radius of 30 µm, 35 µm, and 40 µm are shown in Fig. 2(a), where the fitness function is defined as:

$$Fitness = T_{0-0} + T_{1-1} + T_{2-2} \tag{4}$$

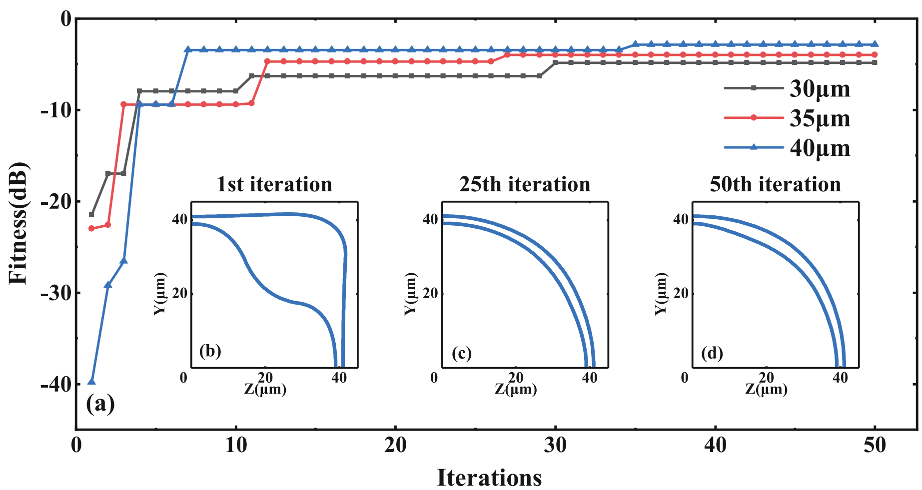

Fig. 2. (a) The EO optimization of the B-spline multimode waveguide bend with radius of 30 µm, 35 µm, and 40 µm, schematic diagrams of the structure in (b) the 1st iteration, (c) the 25th iteration, and (d) the 50th iteration.

T_{0-0}, T_{1-1}, and T_{2-2} represent the transmission of the B-spline multimode waveguide bends for the TE_0, TE_1, and TE_2 modes, respectively. Figures 2(b-d). respectively present schematic diagrams of the B-spline waveguide bend structures with a radius of 40 µm. Figure 1(d). shows a structural comparison of four commonly used waveguide bend curves under the same radius. Using the finite-difference time-domain (FDTD) method, Table 1. compares the insertion losses for TE_0, TE_1, and TE_2 modes across four types of curves: Arc, Euler, Bézier, and B-spline, for radius of 30 µm, 35 µm, and 40 µm. It is evident that for effective radius of 30 µm,35 µm, and 40 µm, B-spline curves exhibit lower insertion losses across all three modes. To compare the structural differences at the best performance for the four curves at the same radius, we have plotted the structural comparison corresponding to the optimal parameters at R_{eff} = 30 µm, 35 µm, and 40 µm, as shown in Fig. 3. Furthermore, we simulated the optimal performance of the four curves as R_{eff} varies from 10 µm to 100 µm, with the results shown in Fig. 4. It can be observed that B-spline curves perform well across different radius, especially for the TE_1 and TE_2 modes. Additionally, it can be inferred that as R_{eff} increases further, the bending performance of the B-spline waveguide multimode bend will improve even more.

With the determined structural parameters $h = 300$ nm, $w = 3.5$ µm and R_{eff} = 100 µm, we simulated and optimized the transmission performance of the multimode waveguide bend. The simulation results in Fig. 5 show that, compared to the traditional arc-shaped waveguide bend, Euler waveguide bend, and Bézier waveguide bend, the optimized B-spline multimode waveguide

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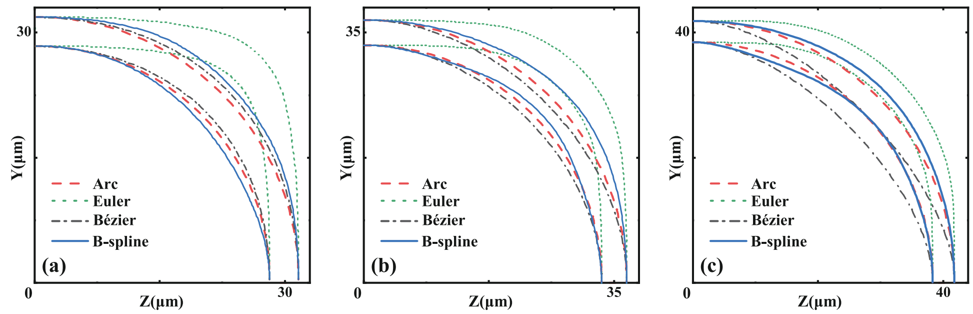

Fig. 3. The structural comparison of the best performance for Arc, Euler, Bézier, and B-spline curves at different R_{eff}, (a) R_{eff} =30 µm, (b) R_{eff} =35 µm, (c) R_{eff} =40 µm.

Table 1. Comparison of simulated bending losses in waveguides with different bend types at radius of 30 µm, 35 µm, and 40 µm, $w = 3.5$ µm

R_{eff}	Mode	Bend type			
		Arc	Euler	Bézier	B-spline
30 µm	TE_0 IL (dB/90°)	1.47	1.19	0.51	0.23
	TE_1 IL (dB/90°)	3.82	3.79	1.27	0.92
	TE_2 IL (dB/90°)	6.97	9.62	7.10	4.36
35 µm	TE_0 IL (dB/90°)	3.57	1.43	0.34	0.14
	TE_1 IL (dB/90°)	6.93	3.24	0.65	0.44
	TE_2 IL (dB/90°)	6.08	8.24	4.79	3.41
40 µm	TE_0 IL (dB/90°)	6.09	2.21	0.21	0.13
	TE_1 IL (dB/90°)	13.36	2.60	0.40	0.48
	TE_2 IL (dB/90°)	8.12	6.91	4.32	2.27

bend exhibits superior performance, with insertion losses for the TE_0, TE_1, and TE_2 modes being lower than 0.04 dB, 0.07 dB, and 0.18 dB, respectively. Additionally, the crosstalk performance is excellent, with the crosstalk for the TE_0, TE_1, and TE_2 modes all being no higher than -26 dB. Figure 6 also compares the mode field distributions of the traditional arc-shaped waveguide, Euler waveguide bend, Bézier waveguide bend, and B-spline multimode waveguide bend for the TE_0, TE_1, and TE_2 modes. The simulation results clearly indicate that, compared to the traditional arc-shaped waveguide, Euler waveguide bend, and Bézier waveguide bend, the B-spline waveguide bend significantly reduces inter-mode crosstalk and mode leakage. Although the Euler

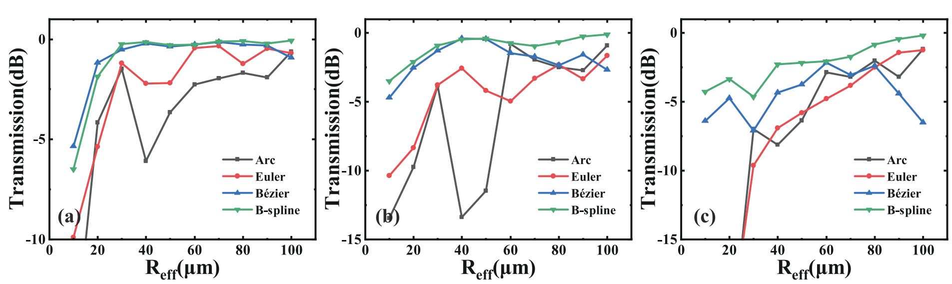

Fig. 4. The simulated performance of the Arc, Euler, Bézier, and B-spline waveguide bends at different radius, (a) TE_0 input, (b) TE_1 input, and (c) TE_2 input.

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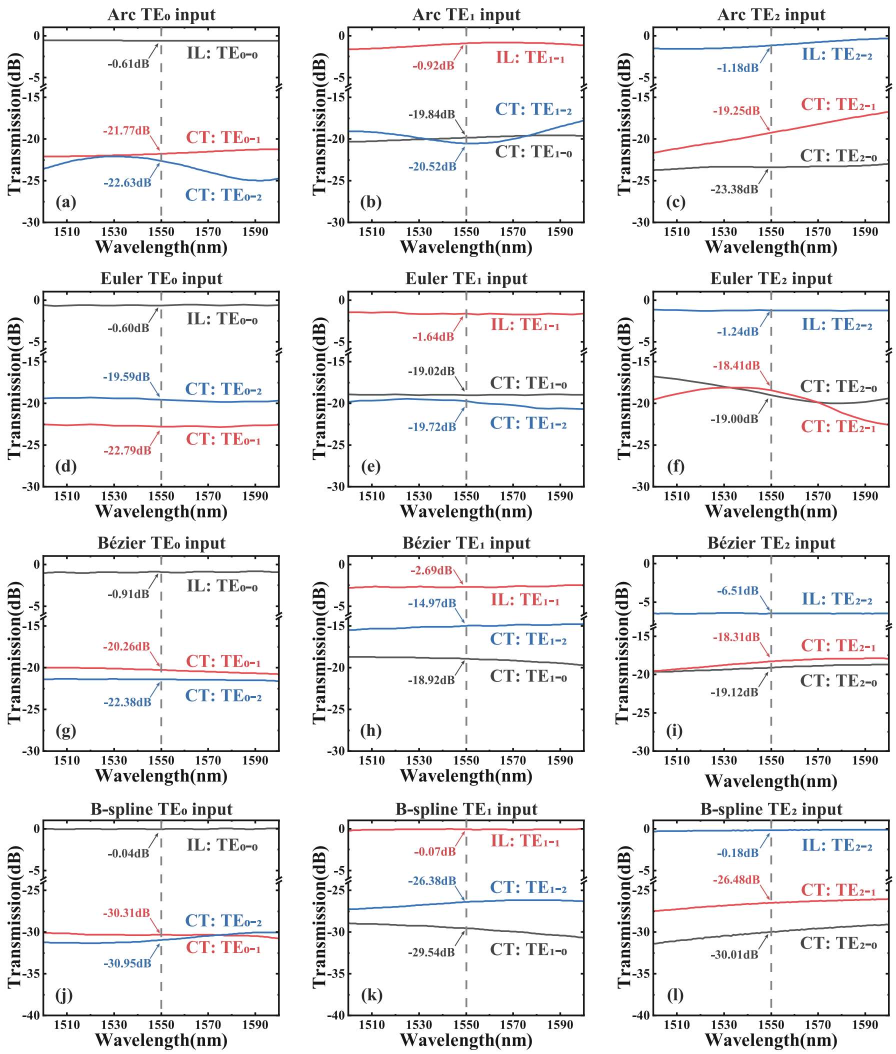

Fig. 5. The simulated transmission spectrum of the traditional arc-shaped waveguide bend with (a) TE_0 input, (b) TE_1 input, (c) TE_2 input, Euler waveguide bend with (d) TE_0 input, (e) TE_1 input, (f) TE_2 input, Bézier waveguide bend with (g) TE_0 input, (h) TE_1 input, (i) TE_2 input, B-spline multimode waveguide bend with (j) TE_0 input, (k) TE_1 input, (l) TE_2 input. IL: insert loss. CT: crosstalk.

and Bézier waveguides perform well in the TE_0 mode, the B-spline waveguide exhibits better performance in the TE_1 and TE_2 modes. Additionally, as shown in this figure, Arc, Euler and Bézier waveguides exhibit relatively low leakage for the TE_0 mode whereas the TE_1 and TE_2 modes suffer from significant leakage. This phenomenon arises from the birefringence of lithium niobatem which causes variations in the effective refractive index during bending, leading to mode mismatch in the curved waveguide. Furthermore, TE-TM mode hybridization may also occur, however noticeable hybridization typically requires a sufficiently long coupling length [55]. Therefore, the primary source of loss can be attributed to mode mismatch caused by the birefringence of lithium niobate during the bending process.

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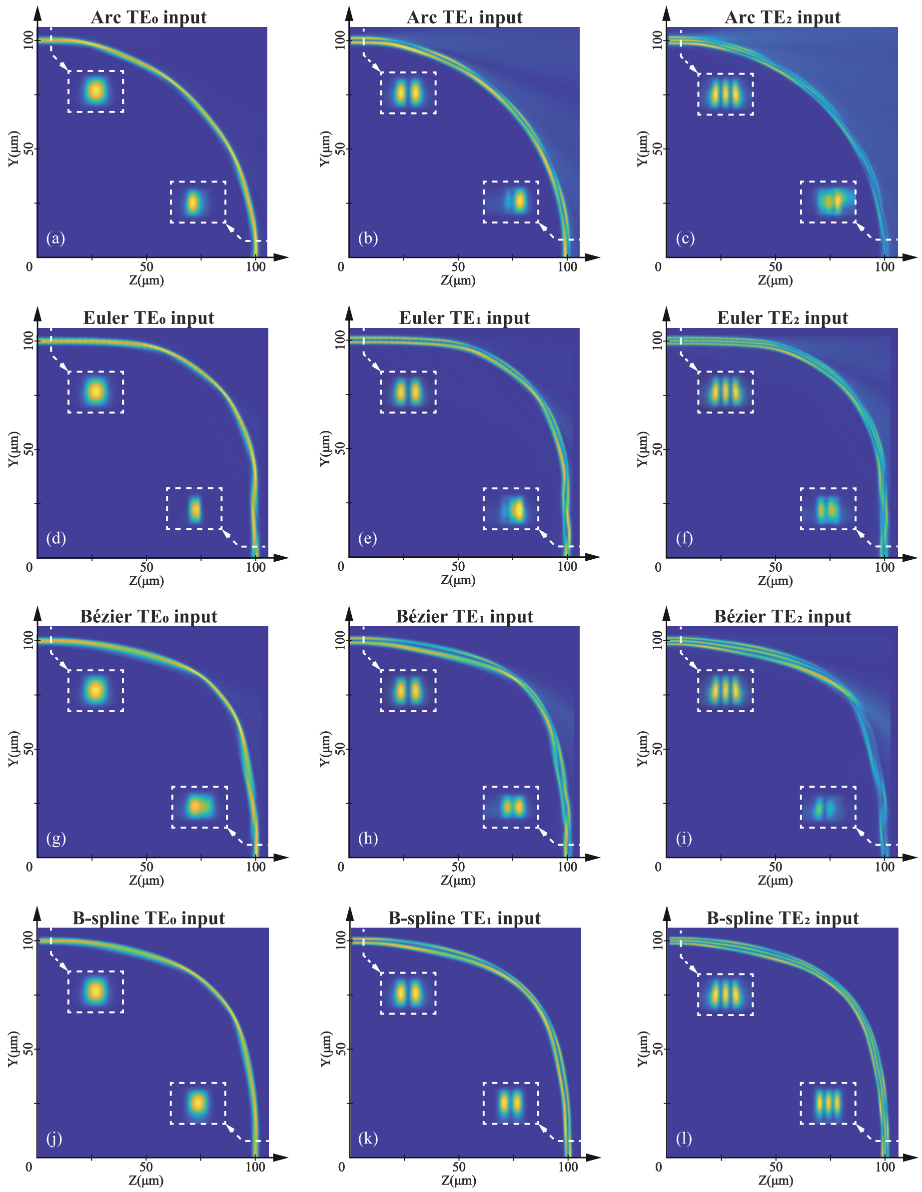

Fig. 6. The simulated field distribution of the traditional arc-shaped waveguide bend with (a) TE_0 input, (b) TE_1 input, (c) TE_2 input, Euler waveguide bend with (d) TE_0 input, (e) TE_1 input, (f) TE_2 input, Bézier waveguide bend with (g) TE_0 input, (h) TE_1 input, (i) TE_2 input, B-spline multimode waveguide bend with (j) TE_0 input, (k) TE_1 input, (l) TE_2 input.

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To further demonstrate the advantages of the width variability of the B-spline multimode waveguide bend mentioned earlier, we analyze the width-fixed Arc-shaped multimode waveguide bend and the B-spline multimode waveguide bend at a 1550 nm as shown in Fig. 7. The comparison shows that, overall, the B-spline multimode waveguide bend exhibits a higher effective refractive index for the same mode. A higher effective refractive index indicates stronger mode confinement, which helps to better confine the optical wave within the waveguide and reduce scattering losses outside the bend.

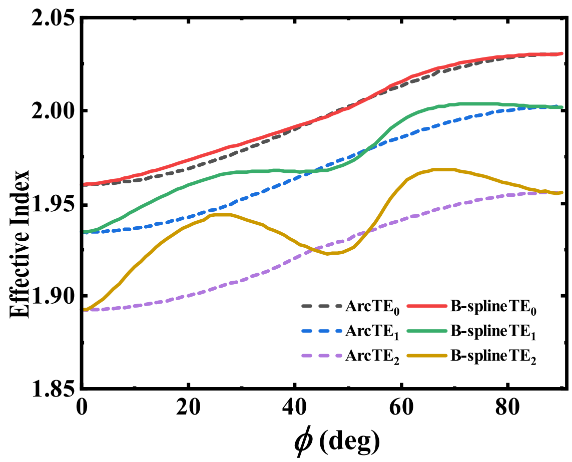

Fig. 7. Effective refractive indices for TE_0, TE_1, and TE_2 modes between the width-fixed arc-shaped waveguide bend and the width-variable B-spline multimode waveguide bend for different ϕ at 1550 nm. The dashed lines represent the width-fixed arc-shaped waveguide, while the solid lines represent the width-variable B-spline waveguide.

3. Results

The device is fabricated on a 600 nm thick X-cut LN film with a 4.7 µm buried silicon oxide layer (NANOLN). The chip fabrication process begins with a thorough cleaning of the LNOI wafer. The wafer undergoes ultrasonic cleaning using acetone, methanol, and isopropanol to remove any organic contaminants from the surface. Next, a layer of CSAR 62 is spin-coated onto the cleaned wafer. The LN waveguides are patterned using electron beam lithography (EBL) and formed through inductively coupled plasma (ICP) etching. The etched waveguides have a depth of 300 nm and a sidewall angle of 70 degrees. Finally, a protective upper cladding of approximately 1 µm of dense silicon oxide is deposited using plasma enhanced chemical vapor deposition (PECVD) at 300°C.

The measurement setup consists of single-mode fibers, a polarization controller, a tunable laser (1525-1575 nm), and a multi-channel optical power meter. As shown in Fig. 8(a), we cascaded different numbers of waveguide bends for linear fitting. To characterize the multimode waveguide bends, we used a pair of 3-channel mode division multiplexers (MDMs) with grating couplers on both sides of the bend, as shown in Fig. 8(b). This setup allows us to excite the modes of interest in the multimode waveguide. The three input ports are labeled I1, I2, and I3, used for coupling the TE_0, TE_1, and TE_2 modes, respectively, and the corresponding output ports are labeled O1, O2, and O3 for measurement.

As shown in Figs. 9(a-c), the results of the two cascaded 90° waveguide bends clearly demonstrate that the B-spline-based waveguide bends have very low insertion loss and mode crosstalk. At a wavelength of 1550 nm, the insertion losses for the TE_0, TE_1, and TE_2 modes in

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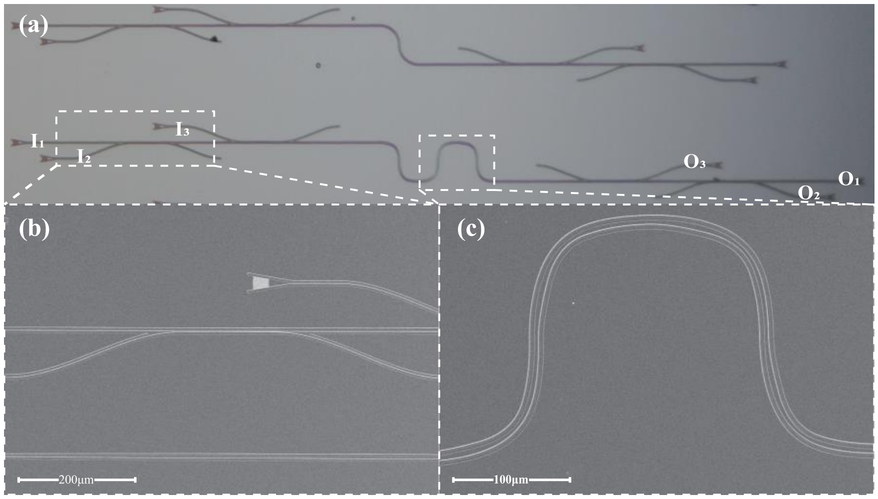

Fig. 8. (a) The microscope image of the fabricated multimode waveguide bend, (b) SEM image of MDM, (c) SEM image of B-spline multimode waveguide bends.

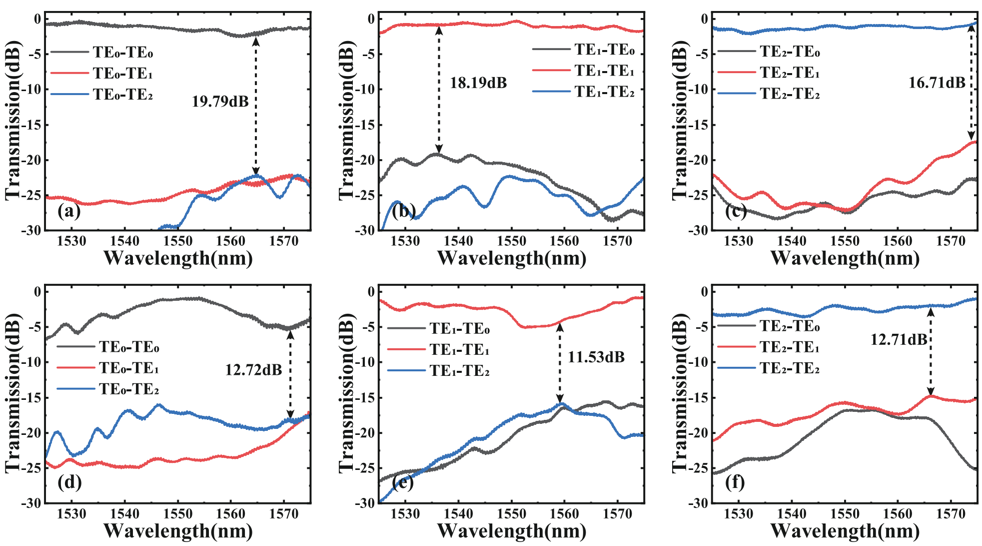

Fig. 9. The measured transmission spectra of two cascaded B-spline multimode waveguide bends for (a) TE_0 input, (b) TE_1 input, and (c) TE_2 input. The transmission measurement diagram of two cascaded traditional arc-shaped multimode waveguide bends for (d) TE_0 input, (e) TE_1 input, and (f) TE_2 input.

the two cascaded B-spline waveguide bends are approximately 1.46 dB, 0.38 dB, and 2.17 dB, respectively, representing a marked improvement over traditional arc-shaped waveguide. To more accurately characterize the performance of multimode curved waveguides, we cascaded different numbers of curved waveguides for linear fitting to eliminate the measurement error introduced by the grating coupler. Specifically, 0, 90, 158, and 226 curved waveguides were cascaded for measuring the transmission in the TE_0 mode, 0, 34, 58, and 110 curved waveguides for the TE_1 mode, and 0, 20, 38, and 62 curved waveguides for the TE_2 mode. Through linear fitting, the insertion losses for the three modes were calculated, with the TE_0, TE_1, and TE_2

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modes having insertion losses of approximately 0.05, 0.10, and 0.29 dB at 1550 nm, respectively, as shown in Fig. 10. For comparison, we fabricated traditional arc-shaped waveguide bends of the same dimensions. The transmission spectra corresponding to different mode inputs are shown in Figs. 9(d-f). Within the 1525 nm to 1575 nm wavelength range, the traditional waveguide bends exhibit significantly higher insertion loss and crosstalk. The performance of the fabricated B-spline waveguide bends is slightly lower than the simulation results, which may be due to fabrication-induced errors in waveguide width and depth.

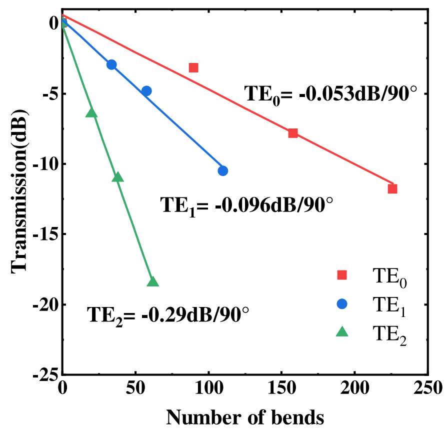

Fig. 10. Measured average insertion losses of the TE_0, TE_1, and TE_2 modes for different numbers of B-spline multimode waveguide bends at 1550 nm.

As shown in Fig. 11, we simulated the device performance with width variations of $\Delta w = \pm 25$ nm and etch depth variations of $\Delta h = \pm 25$ nm. The results indicate that, even with these dimensional discrepancies, the insertion loss remains almost unchanged, with only a slight increase in crosstalk. However, we also observe that within the 1500-1600 nm wavelength range, the simulated crosstalk remains below -18 dB, demonstrating that the device can tolerate fabrication errors.

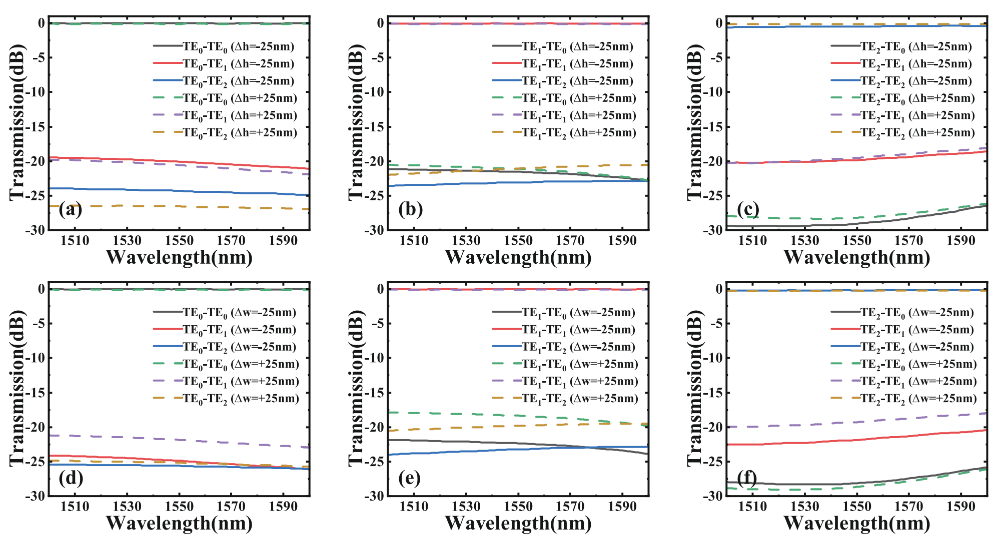

Fig. 11. The simulated transmission spectrum of the B-spline multimode waveguide bend with (a-c) $\Delta h = \pm 25$ nm, (d-f) $\Delta w = \pm 25$ nm.

4. Conclusion

In summary, we proposed and experimentally demonstrated a multimode waveguide bend supporting three modes, designed based on a B-spline curve. This multimode waveguide bend, with an effective radius of 100 μm, a waveguide width of 3.5 μm, a ridge height of 300 nm, a slab thickness of 300 nm, and a sidewall angle $\theta \approx 70°$, achieved an insertion loss of less than 0.29 dB and crosstalk below -16.71 dB for two cascaded bends. Compared to traditional arc-shaped waveguides, this high-performance multimode waveguide bend significantly reduces the bend radius while maintaining similar performance, thereby enhancing the design flexibility of on-chip integration. This structure provides an effective solution for improving the integration density of LNOI-based photonic circuits. Additionally, the B-spline curve, as a smooth and highly flexible curve, can be applied not only to the waveguide bends discussed in this paper but also to the design of various other passive devices, such as waveguide crossings, edge couplers, MDMs, polarization beam splitters (PBSs), add-drop optical filters, and microring resonators. It offers new insights for the design of high-performance on-chip devices.

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Disclosures. The authors declare no conflict of interest.

Data availability. The data that support the findings of this study are available from the corresponding author upon reasonable request.

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